

# NASRUL HUDA

MSc Data Science and Artificial Intelligence

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## PROFESSIONAL SUMMARY

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Data Science and AI student with hands-on experience in machine learning, full-stack development, and research. Passionate about building intelligent systems that solve real-world problems. Strong foundation in Python, modern web technologies, and cloud infrastructure. Seeking opportunities to apply AI/ML expertise in innovative projects.

## EDUCATION

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### University of Hamburg

*Master of Science in Data Science and Artificial Intelligence*

Hamburg, Germany

*Oct 2024 - Present*

### Jamia Millia Islamia

*Bachelor of Technology in Computer Science Engineering*

New Delhi, India

*Oct 2020 - Jun 2024*

## EXPERIENCE

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### SynTwin GmbH

*AI Engineer*

Munich, Germany

*Oct 2025 - Apr 2026, Part Time*

- Developed agentic AI workflows and orchestration pipelines for a Digital Twin platform, designing multi-turn conversational agent architectures using Python and LiveKit.
- Applied prompt engineering strategies to improve agent reliability and consistency across complex, stateful interactions.
- Built and maintained React/Vite frontend components integrated with Supabase for real-time data sync and user authentication.
- Collaborated closely with the CTO and a team of 5 developers on end-to-end system design spanning AI agent architecture, backend services, and user-facing interfaces.
- **Technical Skills:** LiveKit, Supabase, React, Docker Compose, STT, TTS, GitHub Actions

### University of Hamburg - Security and Distributed Systems Research Group

*Research Assistant*

Hamburg, Germany

*July 2025 - Sept 2025, Part Time*

- Built Python-based microservices for signing and verifying cryptographic hash signatures as part of research data management and security analysis tooling.
- Contributed to the NFDIxCS (National Research Data Infrastructure for Computer Science) project, working at the intersection of AI, data science, and distributed systems.
- **Technical Skills:** Flask, ECDSA, RSA, Docker, GitLab CI, Linux, Cybersecurity, Distributed Systems

### Pettoo UG

*Freelance Software Engineer*

Hamburg, Germany

*June 2025 - September 2025, Part Time*

- Designed and implemented the backend of a pet services platform enabling users to discover and book services, purchase products, and connect with other pet owners, using FastAPI and PostgreSQL.
- Architected the database schema and core business logic, and integrated Elasticsearch for search functionality.
- Led a team of 3 developers, coordinating tasks and maintaining consistent code quality across the project.
- Set up CI/CD pipelines with GitHub Actions and deployed containerised services on Google Cloud Platform.
- **Technical Skills:** FastAPI, PostgreSQL, Google Cloud Platform, Docker, Elasticsearch, GitHub Actions

### University of Hamburg - Language Technology Research Group

*Research Assistant*

Hamburg, Germany

*Feb 2025 - May 2025, Part Time*

- Contributed to the DATS Project (Discourse Analysis Tool Suite), an NLP platform for automated text analysis and discourse processing, implementing key features and functionality enhancements.
- Developed full-stack components including a React TypeScript frontend and FastAPI backend services, with RESTful APIs and database integrations.
- Implemented NLP processing modules, text preprocessing pipelines, and data analysis algorithms to handle large-scale linguistic datasets.
- **Technical Skills:** FastAPI, React, NLP, Machine Learning, Docker, Git

- Developed and maintained REST APIs using FastAPI for order processing, validation, and transaction workflows, with database schema design and migrations on Supabase (PostgreSQL) for scalable data management.
- Built a Wix order synchronisation module for automated customer order fetching, integrated Zoho Books API for invoice generation and financial workflow automation, and WhatsApp API for automated notifications.
- Developed a real-time messaging system using WebSockets to improve system responsiveness and user interaction, and collaborated with the team to debug issues, optimise performance, and enhance system reliability.
- **Technical Skills:** FastAPI, Supabase, PostgreSQL, WebSockets, Zoho Books API, WhatsApp API

## PROJECTS

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### **Few-Shot Bioacoustic Sound Event Detection 🦻** | TorchAudio, PyTorch, PyTorch Lightning, Hydra, MLflow

- Developed a few-shot meta-learning system to classify animal vocalizations from as few as 5 labeled examples per species, using episodic (N-way, K-shot) training with Prototypical Networks.
- Engineered audio features from raw .wav files into 128-bin log-mel spectrograms and PCEN arrays using librosa, pre-computed and stored as .npy for efficient training.
- Designed and compared four encoder architectures: ResNet baseline (V1), ResNet with channel & temporal attention + learnable distance metric (V2), Audio Spectrogram Transformer with multi-head self-attention (V3), and pretrained EfficientNet-B1 (V4).
- Built end-to-end training infrastructure with PyTorch Lightning, Hydra configuration management, MLflow experiment tracking, and a custom CLI for training, evaluation, and feature export.

### **Federated Fine-Tuning of Nicheformer 🦻** | Python, PyTorch, Federated Learning

- Implemented federated learning framework for fine-tuning Nicheformer transformer model on mouse brain spatial transcriptomics data, comparing centralized (94.45%), federated (92.76%), and local training strategies
- Built data pipeline for anatomical non-IID partitioning (dorsal/mid/ventral regions) with 24-class cell type classification on 250-dimensional features (248 genes + spatial coordinates) achieving competitive 92.76% accuracy in federated setting

### **GitGood: AI-Powered GitHub Repository Analysis Tool 🦻** | FastAPI, React, Docker, OpenAI API

- Built a full-stack web application that integrates with GitHub OAuth to let users import repositories and interact with their codebase through an AI-powered chat interface.
- Designed a FastAPI backend with SQLAlchemy for user session management and repository data storage, connected to the OpenAI API to answer natural language questions about commit history, diffs, and code evolution.
- Developed a responsive React/TypeScript frontend with Vite and TailwindCSS, with Axios handling API communication between client and server.
- Containerised the full stack using Docker and Docker Compose for consistent local development and production deployment.

### **ACL Anthology Semantic Retrieval System 🦻** | FastAPI, Langchain, React, Qdrant, Fireworks AI, Groq

- Built a Retrieval-Augmented Generation (RAG) system over the ACL Anthology corpus to enable semantic search across tens of thousands of NLP research papers, going beyond keyword matching to understand query meaning using dense vector embeddings.
- Implemented an offline ingestion pipeline that fetches paper metadata and abstracts, generates dense embeddings via Fireworks AI, and indexes them into a Qdrant vector store for fast approximate nearest-neighbour retrieval.
- Designed two query modes: natural language input (reformulated into multiple semantic queries by an LLM via Groq) and "Paper as Query" (using a paper's abstract as a semantic proxy to find related work); results aggregated using Reciprocal Rank Fusion (RRF).
- Developed a FastAPI backend and React frontend with real-time streaming responses, inline citations with hover previews, and a query monitoring panel showing pipeline steps and timing.

### **MyTorch: PyTorch-Inspired Deep Learning Framework 🦻** | NumPy, Deep Learning

- Built a minimal deep learning library from scratch inspired by PyTorch's API, implementing core tensor operations (arithmetic, matrix multiplication, transpose) with a custom automatic differentiation engine and backward pass for gradient computation.
- Implemented neural network modules including fully connected (Linear) layers, ReLU activations, and Cross Entropy loss, all with manual forward and backward propagation with no ML framework dependencies.
- Developed SGD and Adam optimisers from first principles, replicating PyTorch's modular optimizer interface.
- Validated the framework on MNIST digit classification, achieving **97.16% test accuracy** (macro F1: 97.15%) using only linear layers, demonstrating correctness of the custom autograd and training pipeline.

**Momentum** | Web Scraping, Content Analysis, Research Tools, Machine Learning, Python

- Developing a research assistant web application inspired by Anara Labs' Anara, designed to streamline academic and technical research workflows.
- Building automated web content extraction and organization systems to gather, process, and categorize research materials from multiple online sources.
- Implementing intelligent summarization algorithms to condense and present research findings in structured, actionable formats for enhanced productivity.

## TECHNICAL SKILLS

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**Languages:** Python, TypeScript, C++, SQL, Rust (Novice)

**Frameworks:** PyTorch, PyTorch Lightning, Scikit-Learn, Pandas, FastAPI, React, LangChain, Docker, Git

**Databases:** PostgreSQL, Qdrant, Supabase, Redis, Elasticsearch

**Cloud/DevOps:** Google Cloud Platform, Docker, CI/CD, GitHub Actions

## LANGUAGES

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**English:** C1 (Advanced) | **Hindi:** Native | **Urdu:** Native | **German:** A2 (Elementary)

## AWARDS & ACHIEVEMENTS

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**Letter of Excellence** – Vice Chancellor of Jamia Millia Islamia (outstanding academic performance).

**IELTS 8.0 Band (C1)** – R: 9.0, L: 8.0, W: 7.5, S: 7.0.

**1st Rank** – Top position in 4th year of Bachelor's program (2024).

**Certificate of Excellence** – Coding Ninjas (Data Structures and Algorithms).